

POWERFACTORY

PowerFactory 2024

Technical Reference

Common Result Variables for Terminals and Elements

Short-Circuit Calculation

PF2024

POWER SYSTEM SOLUTIONS

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Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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1 General Description

This document describes common variables available for terminals, single- and multiple-port elements following a short-circuit calculation. As a general rule, variable names starting with a capital letter express physical quantities and variable names starting with a lower case letter express per unit values.

1.1 Derived Short-Circuit Currents

Most of the derived short-circuit currents, such as the peak short-circuit current or the breaking current, depend on R/X ratios, either explicitly or implicitly. The calculated R/X ratios are only valid for the short-circuit location due to limitations of the used methodology. Therefore the derived currents, and specifically the branch contributions, can only be calculated for branches directly connected to the short-circuit location.

1.2 Multiple Short-Circuit Locations

Most of the short-circuit calculation methods offer the possibility to process multiple short-circuit location simultaneously. In these cases, the *Result Variables for Terminals* described below are calculated only for the selected short-circuit locations. The *Result Variables for Elements* are calculated for the side connected to a short-circuit location. For example, if both ends of a line are connected to short-circuit locations, the result at each end will reflect the contribution for a short-circuit at that end.

1.3 Symmetrical Component Transformation

Currents, voltages and impedances are commonly provided as phase quantities and as symmetrical components.

Sequence currents or voltages, such as those shown in Table 4.8, are calculated using the following transformation:

$$\mathbf{I}_{012} = \mathbf{T}_{012} \cdot \mathbf{I} \tag{1}$$

where $\mathbf{I}$ is the vector of phase currents and $\mathbf{T}_{012}$ is defined as:

$$\mathbf{T}_{012} = \frac{1}{3} \cdot \begin{bmatrix} 1+0j & 1+0j & 1+0j \\ 1+0j & -0.5+0.5\cdot\sqrt{3}j & -0.5-0.5\cdot\sqrt{3}j \\ 1+0j & -0.5-0.5\cdot\sqrt{3}j & -0.5+0.5\cdot\sqrt{3}j \end{bmatrix}$$

Sequence impedances, such as those shown in Table 4.9, are calculated using the following transformation:

$$\mathbf{Z}_{012} = \mathbf{T}_{012} \cdot \mathbf{Z}_{ABC} \cdot \mathbf{T}_{ABC} \tag{2}$$

where T_{012} is as defined above, Z_{ABC} is the phase impedance matrix and T_{ABC} is:

$$\mathbf{T}_{ABC} = \begin{bmatrix} 1+0j & 1+0j & 1+0j \\ 1+0j & -0.5-0.5\cdot\sqrt{3}j & -0.5+0.5\cdot\sqrt{3}j \\ 1+0j & -0.5+0.5\cdot\sqrt{3}j & -0.5-0.5\cdot\sqrt{3}j \end{bmatrix}$$

2 VDE/IEC Short-Circuit

Short-circuit calculations according to IEC60909 ([2, 3]) differentiate between *near-to-generator* and *far-from-generator* faults in their equations. A short-circuit is considered as *near-to-generator* if any synchronous machine contributes more than twice its nominal current to a fault, or asynchronous motors contribute more than 5% of the short-circuit current.

In addition, differentiation is made between short-circuits in *meshed networks* and *single- or multiple single-fed* short-circuits. A *meshed network* is assumed if any contributing branch is connected to at least two machines or if there are parallel impedances.

The two conditions explained above are evaluated separately for each short-circuit location. There are no variables available that explicitly show the result of the evaluation itself.

2.1 Balanced Representation

2.1.1 Result Variables for Terminals

2.1.1.1 Short-Circuit Current and Derived Values

Table 2.1: Variables for short-circuit current and derived values at terminals

Name	Unit	Description
Ikss	kA	Initial Short-Circuit Current
I	kA	Short-Circuit Current
phii	deg	Phase Current, Angle
Skss	MVA	Initial Short-Circuit Power
ip	kA	Peak Short-Circuit Current
Ib	kA	Short-Circuit Breaking Current
Sb	MVA	Short-Circuit Breaking Power
Ik	kA	Steady-State Short-Circuit Current
Ith	kA	Thermal equivalent Short-Circuit Current
idc	kA	Decaying Component
Iasy	kA	Asymmetrical Breaking Current

The calculated initial short-circuit current, $\underline{\mathbf{I}}_{\mathbf{k}}^{''}$, is available as:

$$\underline{I}_k^{''} = I_k^{''} \cdot e^{j \cdot \varphi_k} \tag{3}$$

where:

$I_k^{''}$ is the magnitude of the initial short-circuit current in kA (*m:Ikss* or *m:I*)

φ_k is the phase angle of the initial short-circuit current in deg (*m:phii*)

The initial short-circuit power, $\mathbf{S_k''}$, is a fictitious value to facilitate comparison of short-circuit strengths across different voltage levels and is calculated as follows:

$$S_k'' = \sqrt{3} \cdot I_k'' \cdot U_N \tag{4}$$

where:

S_k'' is the initial short-circuit power in MVA (*m:Skss*)

I_k'' is the magnitude of the initial short-circuit current in kA (*m:Ikss* or *m:I*)

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

The peak short-circuit current, $\mathbf{i_p}$, is calculated as follows:

- (multiple) single-fed

$$i_p = \sum i_{p,j} \tag{5}$$

- meshed network

$$i_p = \sqrt{2} \cdot \kappa \cdot I_{rot}'' + \sqrt{2} \cdot I_{cdp}'' \tag{6}$$

where:

i_p is the peak short-circuit current in kA (*m:ip*)

$i_{p,j}$ is the peak short-circuit current contribution of the j-th branch in kA (see Section 2.1.2)

κ is a factor depending on the R/X ratio (*m:RtoX*) at the short-circuit location

I_{rot}'' is the contribution of rotating machines to the short-circuit current in kA

I_{cdp}'' is the contribution of full size converters to the short-circuit current in kA

Please refer to the corresponding IEC60909 standard for detailed information regarding the calculation of κ. Short-circuit calculations according to IEC60909-2001 and earlier do not support full size converters and thus always assume $I_{k,cdp}'' = 0$.

The symmetrical breaking current, $\mathbf{I_b}$, is calculated as follows:

	near-to-generator	far-from-generator
(multiple) single-fed	$I_b = \sum I_{b,j}$	$I_b = I_k^{''}$
meshed network	$I_b = \left\| \underline{I}_k^{''} - \sum \frac{\sqrt{3} \cdot \underline{Z}_j \cdot \underline{I}_{k,j}^{''}}{c \cdot U_N} \cdot (1 - \mu_j) \cdot \underline{I}_{k,j}^{''} \right\|$	$I_b = I_k^{''}$

where:

I_b is the symmetrical breaking current in kA (*m:Ib*)

$I_{b,j}$ is the symmetrical breaking current contribution of the j-th branch in kA (see Section 2.1.2)

$\underline{I}_k^{''}$ is the complex initial short-circuit current in kA

$\underline{Z}_j$ is the complex impedance of the j-th machine contributing to the fault, including correction factors, in Ω

$\underline{I}_{k,j}^{''}$ is the complex initial short-circuit current contribution of the j-th machine in kA

μ is the factor for the calculation of the breaking current contribution of the j-th machine

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

The equation for short-circuits in meshed networks is simplified for clarity. Please refer to the corresponding IEC60909 standard for the detailed equation. Short-circuit calculations according to IEC60909-2001 and earlier only use the reactive part of $\underline{Z}_j$, whereas later versions use the full complex impedance.

Similar to the initial short-circuit power, the short-circuit breaking power, $\mathbf{S_b}$, is a fictitious value to facilitate comparison of short-circuit breaking requirements across different voltage levels and is calculated as follows:

$$S_b = \sqrt{3} \cdot I_b \cdot U_N \tag{7}$$

where:

S_b is the short-circuit breaking power in MVA (*m:Sb*)

I_b is the short-circuit breaking current in kA (*m:Ib*)

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

According to IEC60909-2016, the steady-state short-circuit current, $\mathbf{I_k}$ is calculated as follows:

	Max. Short-Circuit Currents	Min. Short-Circuit Currents
(multiple) single-fed	$I_k = \sum I_{k,j}$	$I_k = \sum I_{k,j}$
meshed network	$I_k = I_{b,MO}$	$I_k = I_k^{''}$

According to IEC60909-2001, the steady-state short-circuit current, $\mathbf{I_k}$ is calculated as follows:

	near-to-generator	far-from-generator
(multiple) single-fed	$I_k = \sum I_{k,j}$	$I_k = I_k^{''}$
meshed network	$I_k = I_{k,MO}^{''}$	$I_k = I_k^{''}$

where:

I_k is the steady-state short-circuit current in kA (*m:Ik*)

$I_{k,j}$ is the steady-state short-circuit current contribution of the j-th branch in kA (see Section 2.1.2)

$I_k^{''}$ is the magnitude of the initial short-circuit current in kA

$I_{b,MO}$ is the symmetrical breaking current without motors in kA

$I_{k,MO}^{''}$ is the magnitude of the initial short-circuit current without motors in kA

When calculating according to IEC60909-2001, the way motors are ignored can be configured in the short-circuit command. Please refer to the *PowerFactory* User Manual for additional information.

The asymmetrical breaking current, $\mathbf{I_{b,asym}}$, is used to assess circuit-breaker requirements and is calculated as follows:

$$I_{b,asym} = \sqrt{I_b^2 + i_{DC}^2} \tag{8}$$

where:

$I_{b,asym}$ is the asymmetrical breaking current in kA (*m:Iasy*)

I_b is the symmetrical breaking current in kA (*m:Ib*)

i_{DC} is the decaying component of the short-circuit current in kA (*m:idc*)

The decaying component of the short-circuit current, $\mathbf{i_{DC}}$ is calculated as follows:

$$i_{DC} = \sqrt{2} \cdot I_{rot}^{''} \cdot e^{-2 \cdot \pi \cdot f_N \cdot T_b \cdot R/X_{DC}} \tag{9}$$

where:

i_{DC} is the decaying component of the short-circuit current in kA (*m:idc*)

$I_{rot}^{''}$ is the magnitude of rotating machine contributions to the initial short-circuit current in kA

f_N is the nominal frequency of the network in Hz

T_b is the breaking time in s (*c:Tb*)

R/X_{DC} is the R/X ratio for the DC component calculation (*m:RtoX_b*)

The breaking time depends on the *Used Break Time* (iBrkTime) setting in the short-circuit command. If set to “global”, the *Break Time* (Ta) entered in the command will be used. For “min. of local” or “local” the smallest non-zero *Break Time* (e:Tb) of all breakers (*ElmCoup*) and fuses (*RelFuse*) connected to the short-circuit location will be used.

The thermal equivalent short-circuit current, $\mathbf{I_{th}}$, is a fictitious current expressing the thermal strain on equipment and is calculated as follows:

$$I_{th} = I_k^{''} \cdot \sqrt{m + n} \tag{10}$$

where:

I_{th} is the thermal equivalent short-circuit current in kA (*m:Ith*)

m is a factor expressing the heat effect of the DC component with

$$m = f\left(\kappa, T_k\right)$$

n is a factor expressing the heat effect of the AC component with

$$n = f\left(\frac{I_k^{''}}{I_k}, T_k\right)$$

$I_k^{''}$ is the magnitude of the initial short-circuit current in kA (*m:Ikss*)

I_k is the steady-state short-circuit current in kA (*m:Ik*)

κ is a factor depending on the R/X ratio (*m:RtoX*) at the short-circuit location

T_k is the *Fault Clearing Time (Ith)* (*Tk*) from the short-circuit command in s

For a detailed description of the factors, $\mathbf{m}$ and $\mathbf{n}$, please refer to the IEC60909.

2.1.1.2 Impedances

Table 2.2: Variables for impedances at terminals

Name	Unit	Description
R	Ω	Short-Circuit Impedance, Real Part
X	Ω	Short-Circuit Impedance, Imaginary Part
Z	Ω	Short-Circuit Impedance, Magnitude
phiz	deg	Short-Circuit Impedance, Angle
rSbase	p.u.	Short-Circuit Impedance, Real Part
xSbase	p.u.	Short-Circuit Impedance, Imaginary Part
zSbase	p.u.	Short-Circuit Impedance, Magnitude
XtoR		X/R ratio (peak)
RtoX		R/X ratio (peak)
XtoR_b		X/R ratio (dc)
RtoX_b		R/X ratio (dc)
fctof_b		Equivalent Frequency fc/f (dc)
Tn	ms	Grid Time Constant

The calculated equivalent impedance, $\underline{\mathbf{Z}}_{\mathbf{k}}$, is available as:

$$\underline{Z}_k = R_k + j \cdot X_k = Z_k \cdot e^{j \cdot \varphi_Z} \tag{11}$$

where:

R_k is the real part of the equivalent impedance in Ω (*m:R*)

X_k is the imaginary part of the equivalent impedance in Ω (*m:X*)

Z_k is the magnitude of the equivalent impedance in Ω (*m:Z*)

φ_Z is the angle of the equivalent impedance in deg (*m:phiz*)

The impedance is also available in per unit values (*m:rSbase*, *m:xSbase* and *m:zBase*). The corresponding base impedance, $\mathbf{Z_B}$, is calculated as follows:

$$Z_B = \frac{U_N^2}{S_{base}} \tag{12}$$

where:

S_{base} is the *Base Apparent Power* (Sbase) from the *Project Settings* in MVA

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

In **meshed networks**, the *R/X ratio* (m:RtoX) and its inverse (m:XtoR) are used in the calculation of the *Peak Short-Circuit Current* (m:ip). The *R/X ratio* is calculated based on the short-circuit command settings, typically using the equivalent frequency method. For more information please refer to the *PowerFactory* User Manual or the IEC60909 standard.

For **(multiple) single-fed** short-circuit locations, the *R/X ratio* at the fault location is not used and simply shows the R/X ratio of the equivalent impedance.

Similarly, the *R/X ratio (dc)* (m:RtoX_b) and its inverse (m:XtoR_b) are used in the calculation of the *Decaying Component* (m:idc) of the short-circuit current for meshed networks. They are not used for (multiple) single-fed short-circuit locations. The *Equivalent Frequency fc/f (dc)* (m:fctof_b) shows the equivalent frequency ratio used for the decaying component calculation. For more information please refer to the *PowerFactory* User Manual or the IEC60909 standard.

Note: Both R/X ratios are typically calculated with an equivalent frequency and thus may differ from the R/X ratio obtained from the equivalent impedance, especially in meshed networks.

The grid time constant, $\mathbf{T_N}$, is calculated as follows:

$$T_N = \frac{X}{R} \cdot \frac{1000}{2 \cdot \pi \cdot f_N} \tag{13}$$

where:

T_N is grid time constant in ms (*m:Tn*)

$\frac{X}{R}$ is peak X/R ratio (*m:XtoR*)

f_N is the nominal frequency of the network in Hz

2.1.1.3 Voltages

Table 2.3: Variables for voltages at terminals

Name	Unit	Description
ur	p.u.	Voltage, Real Part
ui	p.u.	Voltage, Imaginary Part
u	p.u.	Voltage, Magnitude
upc	p.u.	Voltage, Magnitude
u1	p.u.	Voltage, Magnitude
u1pc	%	Voltage, Magnitude
u1r	p.u.	Voltage, Magnitude
u1i	p.u.	Voltage, Magnitude
U	kV	Line-Ground Voltage, Magnitude
Ul	kV	Line-Line Voltage, Magnitude
phiu	deg	Voltage, Angle
du	%	Voltage Deviation
urpref	p.u.	Pre-fault Voltage, Real Part
uipref	p.u.	Pre-fault Voltage, Imaginary Part
upref	p.u.	Pre-fault Voltage, Magnitude
uprefpc	%	Pre-fault Voltage, Magnitude
Upref	kV	Line-Ground Pre-fault Voltage, Magnitude
Ulpref	kV	Line-Line Pre-fault Voltage, Magnitude
phiupref	deg	Pre-fault Voltage, Angle
phiui		Angle between Voltage and Current
cfac		Voltage Factor

The calculated subtransient short-circuit voltage, $\underline{\mathbf{u}}''$, is available as:

$$\underline{u}'' = u_r'' + j \cdot u_i'' = u'' \cdot e^{j \cdot \varphi_{u''}} \tag{14}$$

where:

u_r'' is the real part of the subtransient short-circuit voltage in p.u. (*m:ur*)

u_i'' is the imaginary part of the subtransient short-circuit voltage in p.u. (*m:ui*)

u'' is the magnitude of the subtransient short-circuit voltage in p.u. (*m:u*)

$\varphi_{u''}$ is the phase angle of the subtransient short-circuit voltage in deg (*m:phiu*)

The positive sequence variables *m:u1*, *m:u1pc*, *m:u1r* and *m:u1i* are made available to to facilitate comparison between balanced and unbalanced calculations via *Comparison of Results*. They are calculated identically to their non positive sequence counterparts.

The angle between voltage and current, $\varphi_{\mathbf{u,i}}$, is calculated as follows:

$$\varphi_{u,i} = \varphi_{u''} - \varphi_k \tag{15}$$

where:

$\varphi_{u,i}$ is the angle between voltage and current in deg (*m:phiui*)

$\varphi_{u''}$ is the phase angle of the subtransient short-circuit voltage in deg (*m:phiu*)

φ_k is the phase angle of the initial short-circuit current in deg (*m:phii*)

The pre-fault voltage, $\underline{\mathbf{u}}_\mathbf{p}$, used for the calculation is available as:

$$\underline{u}_p = \frac{c \cdot U_N}{\sqrt{3}} = u_{p,r} + j \cdot u_{p,i} = u_p \cdot e^{j \cdot \varphi_{u_p}} \tag{16}$$

where:

$u_{p,r}$ is the real part of the pre-fault short-circuit voltage in p.u. (*m:urpref*)

$u_{p,i}$ is the imaginary part of the pre-fault short-circuit voltage in p.u. (*m:uipref*)

u_p is the magnitude of the pre-fault short-circuit voltage in p.u. (*m:upref*)

φ_{u_p} is the phase angle of the short-circuit voltage in deg (*m:phiupref*)

c is the voltage factor according to the short-command configuration (*m:cfac*)

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

While the pre-fault voltage is always a purely real number in VDE/IEC short-circuit calculations, it is shown as complex number for consistency with other short-circuit calculation methods.

Note: The pre-fault voltage is calculated with the voltage factor configured in the short-circuit command and does not include the additional voltage from converter driven plants.

All voltage variables are also available as denormalised quantities, $\mathbf{U}$ and $\mathbf{U_l}$, which are calculated as follows:

$$U = u \cdot \frac{U_N}{\sqrt{3}} \tag{17}$$

$$U_l = u \cdot U_N \tag{18}$$

where:

U is the magnitude of the line-ground voltage in kV (*m:U*, *m:Upref*)

U_l is the magnitude of the line-line voltage in kV (*m:Ul*, *m:Ulpref*)

u is the magnitude of the voltage in p.u. (*m:u*, *m:upref*)

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

Similarly, the convenience variables *m:upc* and *m:uprefpc* express the magnitude of the corresponding voltage in percent. The variable *m:du* shows the deviation of the short-circuit voltage from the nominal voltage in p.u.

2.1.1.4 Loadings

Table 2.4: Variables for loadings at terminals

Name	Unit	Description
Ithload	%	Loading, Thermal equivalent Current
Ipload	%	Loading, Peak Short-Circuit Current
Ithrtk	kA	Rated Short-Time Current (Tk)

Loading variables require rated values. These can be entered in the *Busbar Type* (TypBar) assigned to the terminal. If no type is assigned or the the corresponding rated value is zero, the loadings are assumed to be zero.

The thermal loading *m:Ithload* is calculated as follows:

$$I_{th,r}(T_k) = I_{th,r}^2 \cdot \frac{T_r}{T_k} \tag{19}$$

$$loading = \frac{I_{th}}{I_{th,r}(T_k)} \tag{20}$$

where:

$I_{th,r}$ is the rated short-time thermal current in kA (*t:Ithlim*)

T_r is the rated short-time in s (*t:Tkr*)

T_k is the *Fault Clearing Time (Ith)* (*Tk*) from the short-circuit command in s

$I_{th,r}(T_k)$ is the short-time thermal current rated to the fault clearing time in kA (*m:Ithrk*)

I_{th} is the calculated thermal equivalent short-circuit current in kA (*m:Ith*)

The peak short-circuit loading *m:lpload* is calculated as follows:

$$loading = \frac{i_p}{I_{p,r}} \tag{21}$$

where:

$I_{p,r}$ is the rated peak short-circuit current in kA (*t:lplim*)

i_p is the calculated peak short-circuit current in kA (*m:ip*)

2.1.2 Result Variables for Elements

Variables described in this section are available “per Connection”, unless stated otherwise. The corresponding designation has been omitted for readability.

2.1.2.1 Short-Circuit Current and Derived Values

Table 2.5: Variables for short-circuit current and derived values at elements

Name	Unit	Description
Ikss	kA	Initial Short-Circuit Current
ikss	p.u.	Initial Short-Circuit Current, Magnitude
phii	deg	Phase Current, Angle
Skss	MVA	Initial Short-Circuit Power
ip	kA	Peak Short-Circuit Current
Ib	kA	Short-Circuit Breaking Current
Sb	MVA	Short-Circuit Breaking Power
Ik	kA	Steady-State Short-Circuit Current
Ith	kA	Thermal equivalent Short-Circuit Current
I	kA	Short-Circuit Current
maxIkss	kA	Max. Initial Short-Circuit Current
maxip	kA	Max. Peak Short-Circuit Current
maxIb	kA	Max. Breaking Current
maxIth	kA	Max. Thermal equivalent Short-Circuit Current
i1	p.u.	Positive-Sequence Current, Magnitude
I1	kA	Positive-Sequence Current, Magnitude
phii1	deg	Positive-Sequence Current, Angle
phiui	deg	Angle between Voltage and Current
phiu1i1	deg	Angle between Voltage and Current in Pos. Sequence System
cosphiui		cosphi

The calculated initial short-circuit contribution, $\underline{\mathbf{I}}''_{\mathbf{k,j}}$, is available as:

$$\underline{I}''_{k,j} = I''_{k,j} \cdot e^{j \cdot \varphi''_{k,j}} = i''_{k,j} \cdot I_r \cdot e^{j \cdot \varphi''_{k,j}} \tag{22}$$

where:

$I''_{k,j}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss* or *m:I*)

$\varphi''_{k,j}$ is the phase angle of the initial short-circuit contribution in deg (*m:phii*)

$i''_{k,j}$ is the magnitude of the initial short-circuit contribution in p.u. (*m:ikss*)

I_r is the rated current of the element in kA

The initial short-circuit power contribution, S''_k, is a fictitious value to facilitate comparison of short-circuit strengths across different voltage levels and is calculated as follows:

$$S''_{k,j} = \sqrt{3} \cdot I''_{k,j} \cdot U_N \tag{23}$$

where:

$S''_{k,j}$ is the initial short-circuit power contribution in MVA (*m:Skss*)

$I''_{k,j}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss* or *m:I*)

U_N is the nominal voltage in kV (*e:uknom* of the connected terminal)

The peak short-circuit current contribution, $\mathbf{i}_{\mathbf{p,j}}$, is calculated as follows:

- (multiple) single-fed
$$i_{p,j} = \sqrt{2} \cdot \kappa_j \cdot I''_{j,rot} + \sqrt{2} \cdot I''_{j,cdp} \tag{24}$$
- meshed network
$$i_{p,j} = \sqrt{2} \cdot \kappa \cdot I''_{j,rot} + \sqrt{2} \cdot I''_{j,cdp} \tag{25}$$

where:

$i_{p,j}$ is the peak short-circuit current contribution in kA (*m:ip*)

κ_j is a factor depending on the R/X ratio of the branch (*m:RtoX*)

κ is a factor depending on the R/X ratio at the short-circuit location (*n:RtoX*)

$I''_{j,rot}$ is the rotating part of the initial short-circuit contribution in kA

$I''_{j,cdp}$ is the full size converter part of the initial short-circuit contribution in kA

Please refer to the corresponding IEC60909 standard for detailed information regarding the calculation of κ. Short-circuit calculations according to IEC60909-2001 and earlier do not support full size converters and thus always assume $I''_{k,cdp} = 0$.

Note: The R/X ratio of the branch considers the complete impedance to the source of the contribution, not just the directly connected element.

The short-circuit breaking current contribution, $\mathbf{I_{b,j}}$, is calculated as follows:

- (multiple) single-fed

$$I_{b,j} = I_{b,source} \tag{26}$$

- meshed network

$$I_{b,j} = I_b \cdot \frac{I''_{k,j}}{I''_k} \tag{27}$$

where:

$I_{b,j}$ is the short-circuit breaking current contribution in kA (*m:Ib*)

$I_{b,source}$ is the short-circuit breaking current contribution of the source in kA

I_b is the symmetrical breaking current at the short-circuit location in kA (*n:Ib*)

$I''_{k,j}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss*)

I''_k is the magnitude of the initial short-circuit current at the short-circuit location in kA (*n:Ikss*)

Please refer to the corresponding IEC60909 standard or the Technical Reference of the source element for detailed information regarding the calculation of the breaking current contribution in (multiple) single-fed scenarios.

Similar to the initial short-circuit power, the short-circuit breaking power contribution, $\mathbf{S_{b,j}}$, is a fictitious value to facilitate comparison of short-circuit breaking requirements across different voltage levels and is calculated as follows:

$$S_{b,j} = \sqrt{3} \cdot I_{b,j} \cdot U_N \tag{28}$$

where:

$S_{b,j}$ is the short-circuit breaking power contribution in MVA (*m:Sb*)

$I_{b,j}$ is the short-circuit breaking current contribution in kA (*m:Ib*)

U_N is the nominal voltage in kV (*e:uknom* of the connected terminal)

The stead-state short-circuit current contribution, $\mathbf{I_{k,j}}$, is calculated as follows:

- (multiple) single-fed

$$I_{k,j} = I_{k,source} \tag{29}$$

- meshed network

$$I_{k,j} = I_k \cdot \frac{I''_{k,j}}{I''_k} \tag{30}$$

where:

$I_{k,j}$ is the stead-state short-circuit current contribution in kA (*m:Ik*)

$I_{k,source}$ is the steady-state short-circuit current contribution of the source in kA

I_k is the steady-state short-circuit current at the short-circuit location in kA (*n:Ik*)

$I_{k,j}^{''}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss*)

$I_k^{''}$ is the magnitude of the initial short-circuit current at the short-circuit location in kA (*n:Ikss*)

Please refer to the corresponding IEC60909 standard or the Technical Reference of the source element for detailed information regarding the calculation of the steady-state short-current contribution in (multiple) single-fed scenarios.

The thermal equivalent short-circuit current contribution, $\mathbf{I_{th,j}}$, is a fictitious current expressing the thermal strain on equipment and is calculated as follows:

- (multiple) single-fed

$$I_{th,j} = I_{k,j}^{''} \cdot \sqrt{m_j + n_j} \tag{31}$$

- meshed network

$$I_{th,j} = I_t h \cdot \frac{I_{k,j}^{''}}{I_k^{''}} \tag{32}$$

where:

$I_{th,j}$ is the thermal equivalent short-circuit current contribution in kA (*m:Ith*)

m_j is a factor expressing the heat effect of the DC component with

$$m_j = f\left(\kappa_j, T_k\right)$$

n_j is a factor expressing the heat effect of the AC component with

$$n_j = f\left(\frac{I_{k,j}^{''}}{I_{k,j}}, T_k\right)$$

$I_{k,j}^{''}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss*)

$I_{k,j}$ is the steady-state short-circuit contribution in kA (*m:Ik*)

κ_j is a factor depending on the R/X ratio of the branch (*m:RtoX*)

T_k is the *Fault Clearing Time (Ith)* (*Tk*) from the short-circuit command in s

I_{th} is the thermal equivalent short-circuit current at the short-circuit location in kA (*n:Ith*)

$I_k^{''}$ is the magnitude of the initial short-circuit current at the short-circuit location in kA (*n:Ikss*)

For a detailed description of the factors, $\mathbf{m}$ and $\mathbf{n}$, please refer to the IEC60909.

The calculation of the maximum short-circuit currents depends on the *Calculate max. Branch Currents = Busbar Currents* (iopt_brc) setting in the short-circuit command. If the setting is *enabled*, the values are taken directly from the short-circuit location. If the setting is *disabled*, the value for $I''_{k,max}$ is calculated as follows:

$$I''_{k,max} = \left| \max \left(\underline{I}''_{k} - \underline{I}''_{k,j}; \underline{I}''_{k,j} \right) \right| \tag{33}$$

where:

$\underline{I}''_{k}$ is the complex initial short-circuit at the terminal in kA

$\underline{I}''_{k,j}$ is the complex initial short-circuit contribution of the branch in kA

The resulting values for $I''_{k,max}$ and the corresponding partial rotating and full size converter parts are then used with the corresponding equations to calculate the derived values $i_{p,max}$, $I_{b,max}$ and $I_{th,max}$. Where required, the R/X ratio from the short-circuit location is used.

The angle between voltage and current, $\varphi_{\mathbf{ui}}$, is calculated as follows:

$$\varphi_{ui} = \varphi_u - \varphi_i \tag{34}$$

where:

φ_{ui} is the angle between voltage and current in deg (*m:phiui*)

φ_u is the voltage angle in deg (*m:phiu* of the connected terminal)

φ_i is the initial short-current angle in deg (*m:phii*)

The positive sequence variables *m:i1*, *m:I1*, *m:phii1*, *m:phiu1i1* are made available to to facilitate comparison between balanced and unbalanced calculations via *Comparison of Results*. They are calculated identically to their non positive sequence counterparts. The variable *m:cosphiui* is a convenience variable, showing the power factor for the calculated angle between voltage and current.

2.1.2.2 Impedances

Table 2.6: Variables for impedances at elements

Name	Unit	Description
R	Ω	Fault Impedance, Real Part
X	Ω	Fault Impedance, Imaginary Part
Z	Ω	Fault Impedance, Magnitude
phiz	deg	Fault Impedance, Angle
rSbase	p.u.	Fault Impedance, Real Part
xSbase	p.u.	Fault Impedance, Imaginary Part
zSbase	p.u.	Fault Impedance, Magnitude
XtoR		X/R ratio
RtoX		R/X ratio
Zs	Ω	Source Impedance, Magnitude
phizs	deg	Source Impedance, Angle
Rs	Ω	Source Impedance, Real Part
Xs	Ω	Source Impedance, Imaginary Part

The fault impedance, $\underline{\mathbf{Z}}$, is calculated as follows:

$$\underline{Z} = R + j \cdot X = Z \cdot e^{j \cdot \varphi_Z} = \frac{\underline{U}''}{\underline{I}_k''} \tag{35}$$

where:

R is the real part of the fault impedance in Ω (*m:R*)

X is the imaginary part of the fault impedance in Ω (*m:X*)

Z is the magnitude of the fault impedance in Ω (*m:Z*)

φ_Z is the angle of the fault impedance in deg (*m:phiz*)

$\underline{U}''$ is the complex subtransient voltage of the connected terminal in kV

$\underline{I}_k''$ is the complex subtransient current in kA

The convenience variables *m:XtoR* and *m:RtoX* show the X/R ratio and R/X ratio of the branch impedance.

Note: The fault impedance is back-calculated from calculation results. It therefore includes additional influences, such as correction factors, and will not necessarily match input values.

The fault impedance is also available in per unit values (*m:rSbase*, *m:xSbase* and *m:zBase*). The corresponding base impedance, $\mathbf{Z_B}$, is calculated as follows:

$$Z_B = \frac{U_N^2}{S_{base}} \tag{36}$$

where:

S_{base} is the *Base Apparent Power* (Sbase) from the *Project Settings* in MVA

U_N is the nominal voltage in kV (*e:uknom* of the connected terminal)

The source impedance, $\underline{\mathbf{Z}}_\mathbf{s}$, is calculated as follows:

$$\underline{Z}_s = R_s + j \cdot X_s = Z_s \cdot e^{j \cdot \varphi_{Z_s}} = \frac{\underline{U}_p - \underline{U}''}{\underline{I}_k''} \tag{37}$$

where:

R_s is the real part of the source impedance in Ω (*m:Rs*)

X_s is the imaginary part of the source impedance in Ω (*m:Xs*)

Z_s is the magnitude of the source impedance in Ω (*m:Zs*)

φ_{Z_s} is the angle of the source impedance in deg (*m:phizs*)

$\underline{U}''$ is the complex subtransient voltage of the connected terminal in kV

$\underline{I}_k''$ is the complex subtransient current in kA

$\underline{U}_p$ is the complex pre-fault voltage of the connected terminal in kV

2.1.2.3 Powers

Table 2.7: Variables for powers at elements

Name	Unit	Description
P	MW	Active Power
Q	Mvar	Reactive Power
i1P	p.u.	Positive-Sequence Active Current
i1Q	p.u.	Positive-Sequence Reactive Current
I1P	kA	Positive-Sequence Active Current
I1Q	kA	Positive-Sequence Reactive Current

The power flows, $\mathbf{P}$ and $\mathbf{Q}$, are calculated as follows:

$$P + j \cdot Q = \sqrt{3} \cdot \underline{U}^{''} \cdot \left(\underline{I}_k^{''}\right)^* \tag{38}$$

where:

P is active power in MW (*m:P*)

Q is reactive power in Mvar (*m:Q*)

$\underline{U}^{''}$ is the complex subtransient voltage of the connected terminal in kV

$\left(\underline{I}_k^{''}\right)^*$ is the complex conjugate of the subtransient current in kA

Note: The initial short-circuit power $S_k^{''}$ is a fictitious value and does not express the magnitude of the apparent power corresponding to P and Q.

The positive sequence active and reactive currents are calculated as follows:

$$\varphi = \angle\left(P + j \cdot Q\right)$$

$$I_{1,P} = i_{1,p} \cdot I_r \cdot \cos\varphi \tag{39}$$

$$I_{1,Q} = i_{1,q} \cdot I_r \cdot \sin\varphi \tag{40}$$

where:

φ is the angle of the apparent power

$I_{1,P}$ is the positive sequence active current in kA (*m:I1P*)

$I_{1,Q}$ is the positive sequence reactive current in kA (*m:I1Q*)

$i_{1,P}$ is the positive sequence active current in p.u. (*m:i1P*)

$i_{1,Q}$ is the positive sequence reactive current in p.u. (*m:i1Q*)

I_r is the rated current of the element in kA

Note: Close to the fault location the apparent power will be almost zero. The angle, and by extension the positive sequence active and reactive currents, will therefore be undefined.

2.1.2.4 Beaker Loadings

Table 2.8: Variables for breaker loadings at elements

Name	Unit	Description
BrkIpload	%	Breaker Loading (peak)
BrkIbload	%	Breaker Loading (breaking)
BrkmaxIb	kA	Max. Breaking Current (breaker time)

Loading variables require rated values. These can be entered in the *Switch Type* (TypSwitch) assigned to the breaker associated with the connection side. If no type is assigned or the the corresponding rated value is zero, the loadings are assumed to be zero.

The peak short-circuit loading *m:BrkIpload* is calculated as follows:

$$loading = \frac{i_{p,max}}{I_{p,r}} \tag{41}$$

where:

$I_{p,r}$ is the rated peak short-circuit current in kA (*t:Ipnom* of the breaker)

$i_{p,max}$ is the calculated maximum peak short-circuit current in kA (*m:maxip*)

The break short-circuit loading *m:BrkIbload* is calculated as follows:

$$loading = \frac{I_{b,max}}{I_{b,r}} \tag{42}$$

where:

$I_{b,r}$ is the rated breaking current in kA (*t:Ibnom* of the breaker)

$I_{b,max}$ is the calculated maximum breaking current in kA (*m:BrkmaxIb*)

The *Max. Breaking Current (breaker time)* (m:BrkmaxIb) is identical to the *Max. Breaking Current* (m:maxIb).

2.1.2.5 Protection Related

Table 2.9: Protection related variables at elements

Name	Unit	Description
Tfct	s	Fault Clearing Time

The fault clearing time, $\mathbf{T_{fct}}$ is calculated as follows:

$$T_{fct} = T_{trip} + T_{open} \tag{43}$$

where:

T_{fct} is the fault clearing time in s (*m:Tfct*)

T_{trip} is the calculated tripping time in s (*c:yout* of the relay (*ElmRelay*) or fuse (*RelFuse*))

T_{open} is the breaker opening time in s (*t:t_open* of the breaker)

If multiple protection operate the same breaker, the fastest tripping time is used. A value of 9999.999 s indicates that no trip occurred.

Note: The fault clearing time is only calculated if protection devices are considered in the short-circuit calculation.

2.2 Unbalanced Representation

2.2.1 Result Variables for Terminals

For most variables, please refer to the definitions provided in Section 2.1.1.

Short-Circuit Current and Derived Values

Table 2.10: Variables for short-circuit current and derived values at terminals

Name	Unit	Description
$Ikss_{A,B,C}$	kA	Initial Short-Circuit Current
$I_{A,B,C}$	kA	Short-Circuit Current
$phii_{A,B,C}$	deg	Phase Current, Angle
$Skss_{A,B,C}$	MVA	Initial Short-Circuit Power
$ip_{A,B,C}$	kA	Peak Short-Circuit Current
$Ib_{A,B,C}$	kA	Short-Circuit Breaking Current
$Sb_{A,B,C}$	MVA	Short-Circuit Breaking Power
$Ik_{A,B,C}$	kA	Steady-State Short-Circuit Current
$Ith_{A,B,C}$	kA	Thermal equivalent Short-Circuit Current
$idc_{A,B,C}$	kA	Decaying Component
$Iasy_{A,B,C}$	kA	Asymmetrical Breaking Current

Most phase variables are calculated in a similar manner to those in Table 2.1, however some minor differences exist. Specifically, the breaking current, I_b, is calculated as follows:

$$I_b = I_k^{''} \tag{44}$$

and similarly, the steady-state short-circuit current, I_k, is:

$$I_k = I_k^{''} \tag{45}$$

2.2.2 Result Variables for Elements

For most variables, please refer to the definitions provided in Section 2.1.2.

3 Complete Short-Circuit

Short-circuit calculation with the *complete* method is based on the Engineering Recommendation G74 [1]. It is typically used with *load flow superposition*.

3.1 Balanced Representation

3.1.1 Result Variables for Terminals

3.1.1.1 Short-Circuit Current and Derived Values

Table 3.1: Variables for short-circuit current and derived values at terminals

Name	Unit	Description
Ikss	kA	Initial Short-Circuit Current
I	kA	Short-Circuit Current
phii	deg	Phase Current, Angle
Skss	MVA	Initial Short-Circuit Power
Iks	kA	Transient Short-Circuit Current
phiiks	deg	Transient Short-Circuit Current, Angle
ip	kA	Peak Short-Circuit Current
Ib	kA	Short-Circuit Breaking Current
Sb	MVA	Short-Circuit Breaking Power
ib	kA	Peak Breaking Current
Ibasym	kA	Asymmetrical Breaking Current
idc	kA	Decaying Component
Ith	kA	Thermal equivalent Short-Circuit Current

The calculated initial short-circuit current, $\underline{\mathbf{I}}_{\mathbf{k}}^{''}$, is available as:

$$\underline{I}_k^{''} = I_k^{''} \cdot e^{j \cdot \varphi_k^{''}} \tag{46}$$

where:

$I_k^{''}$ is the magnitude of the initial short-circuit current in kA (*m:Ikss* or *m:I*)

$\varphi_k^{''}$ is the phase angle of the initial short-circuit current in deg (*m:phii*)

The initial short-circuit power, $S_k^{''}$, is a fictitious value to facilitate comparison of short-circuit strengths across different voltage levels and is calculated as follows:

$$S_k^{''} = \sqrt{3} \cdot I_k^{''} \cdot U_N \tag{47}$$

where:

$S_k^{''}$ is the initial short-circuit power in MVA (*m:Skss*)

$I_k^{''}$ is the magnitude of the initial short-circuit current in kA (*m:Ikss* or *m:I*)

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

The calculated transient short-circuit current, $\underline{\mathbf{I}}_{\mathbf{k}}^{'}$, is available as:

$$\underline{I}_k^{'} = I_k^{'} \cdot e^{j \cdot \varphi_k^{'}} \tag{48}$$

where:

$I_k^{'}$ is the magnitude of the transient short-circuit current in kA (*m:Iks*)

$\varphi_k^{'}$ is the phase angle of the transient short-circuit current in deg (*m:phiiks*)

The peak short-circuit current, $\mathbf{i_p}$, is calculated as follows:

$$i_p = \sqrt{2} \cdot \kappa \cdot I_{rot}^{''} + \sqrt{2} \cdot I_{cdp}^{''} \tag{49}$$

where:

i_p is the peak short-circuit current in kA (*m:ip*)

κ is a factor depending on the R/X ratio (*m:RtoX*) at the short-circuit location

$I_{rot}^{''}$ is the magnitude of rotating machine contributions to the initial short-circuit current in kA

$I_{cdp}^{''}$ is the magnitude of full size converter contributions to the initial short-circuit current in kA

The short-circuit breaking current, $\mathbf{I_b}$, is calculated as follows:

$$I_b = \begin{cases} I_k^{'} + (I_{rot,break}^{''} - I_{rot}^{'}) \cdot e^{-25 \cdot T_b} & T_b < 120 \text{ ms} \\ I_k^{'} & T_b \geq 120 \text{ ms} \end{cases} \tag{50}$$

where:

I_b is the short-circuit breaking current in kA (*m:Ib*)

$I_{rot,break}^{''}$ is the magnitude of rotating machine contributions (excluding DFIGs) to the initial short-circuit current in kA

$I_{rot}^{'}$ is the magnitude of rotating machine contributions to the transient short-circuit current in kA

$I_k^{'}$ is the magnitude of the transient short-circuit current in kA (*m:Iks*)

T_b is the breaking time in s (*c:Tb*)

The breaking time depends on the *Used Break Time* (iBrkTime) setting in the short-circuit command. If set to “global”, the *Break Time* (Ta) entered in the command will be used. For “min. of local” or “local” the smallest non-zero *Break Time* (e:Tb) of all breakers (*ElmCoup*) and fuses (*RelFuse*) connected to the short-circuit location will be used.

Similar to the initial short-circuit power, the short-circuit breaking power, $\mathbf{S_b}$, is a fictitious value to facilitate comparison of short-circuit breaking requirements across different voltage levels and is calculated as follows:

$$S_b = \sqrt{3} \cdot I_b \cdot U_N \tag{51}$$

where:

S_b is the short-circuit breaking power in MVA (*m:Sb*)

I_b is the short-circuit breaking current in kA (*m:Ib*)

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

The peak breaking current, $\mathbf{i_b}$, and the asymmetrical breaking current, $\mathbf{I_{b,asym}}$, are used to assess circuit-breaker requirements and are calculated as follows:

$$i_b = \sqrt{2} \cdot I_b + i_{DC} \tag{52}$$

$$I_{b,asym} = \sqrt{I_b^2 + i_{DC}^2} \tag{53}$$

where:

i_b is the peak breaking current in kA (*m:ib*)

$I_{b,asym}$ is the asymmetrical breaking current in kA (*m:Ibasym*)

I_b is the short-circuit breaking current in kA (*m:Ib*)

i_{DC} is the decaying component of the short-circuit current in kA (*m:idc*)

The decaying component of the short-circuit current, $\mathbf{i_{DC}}$ is calculated as follows:

$$i_{DC} = \sqrt{2} \cdot I_{rot}^{''} \cdot e^{-2 \cdot \pi \cdot f_N \cdot T_b \cdot R/X_{DC}} \tag{54}$$

where:

i_{DC} is the decaying component of the short-circuit current in kA (*m:idc*)

$I_{rot}^{''}$ is the magnitude of rotating machine contributions to the initial short-circuit current in kA

f_N is the nominal frequency of the network in Hz

T_b is the breaking time in s (*c:Tb*)

R/X_{DC} is the R/X ratio for DC component calculation (*m:RtoX_b*)

The thermal equivalent short-circuit current, $\mathbf{I_{th}}$, is a fictitious current expressing the thermal strain on equipment and is calculated as follows:

$$I_{th} = I_k^{''} \cdot \sqrt{m+n} \tag{55}$$

where:

I_{th} is the thermal equivalent short-circuit current in kA (*m:Ith*)

m is a factor expressing the heat effect of the DC component with

$$m = f\left(\kappa, T_k\right)$$

n is a factor expressing the heat effect of the AC component with

$$n = f\left(\frac{I_k^{''}}{I_k^{'}}, T_k\right)$$

$I_k^{''}$ is the magnitude of the initial short-circuit current in kA (*m:Ikss*)

$I_k^{'}$ is the magnitude of the transient short-circuit current in kA (*m:Iks*)

κ is a factor depending on the R/X ratio (*m:RtoX*) at the short-circuit location

T_k is the *Fault Clearing Time (Ith)* (*Tk*) from the short-circuit command in s

For a detailed description of the factors, $\mathbf{m}$ and $\mathbf{n}$, please refer to the IEC60909 [2, 3]. Due to the absence of a steady-state current, the *complete* method uses the transient short-circuit current (*m:Iks*) for calculating the factor n.

3.1.1.2 Impedances

Table 3.2: Variables for impedances at terminals

Name	Unit	Description
R	Ω	Short-Circuit Impedance, Real Part
X	Ω	Short-Circuit Impedance, Imaginary Part
Z	Ω	Short-Circuit Impedance, Magnitude
phiz	deg	Short-Circuit Impedance, Angle
rSbase	p.u.	Short-Circuit Impedance, Real Part
xSbase	p.u.	Short-Circuit Impedance, Imaginary Part
zSbase	p.u.	Short-Circuit Impedance, Magnitude
XtoR		X/R ratio (peak)
RtoX		R/X ratio (peak)
XtoR_b		X/R ratio (dc)
RtoX_b		R/X ratio (dc)
fctof_b		Equivalent Frequency fc/f (dc)
Tn	ms	Grid Time Constant

The calculated equivalent impedance, $\underline{\mathbf{Z}}_{\mathbf{k}}$, is available as:

$$\underline{Z}_k = R_k + j \cdot X_k = Z_k \cdot e^{j \cdot \varphi_Z} \tag{56}$$

where:

R_k is the real part of the equivalent impedance in Ω (*m:R*)

X_k is the imaginary part of the equivalent impedance in Ω (*m:X*)

Z_k is the magnitude of the equivalent impedance in Ω (*m:Z*)

φ_Z is the angle of the equivalent impedance in deg (*m:phiz*)

The impedance is also available in per unit values (*m:rSbase*, *m:xSbase* and *m:zBase*). The corresponding base impedance, $\mathbf{Z_B}$, is calculated as follows:

$$Z_B = \frac{U_N^2}{S_{base}} \tag{57}$$

where:

S_{base} is the *Base Apparent Power* (Sbase) from the *Project Settings* in MVA

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

The *R/X ratio* (m:RtoX) and its inverse (m:XtoR) are used in the calculation of the *Peak Short-Circuit Current* (m:ip). The *R/X ratio* is calculated based on the short-circuit command settings, typically using the equivalent frequency method. For more information please refer to the *PowerFactory* User Manual or the IEC60909 standard.

Similarly, the *R/X ratio (dc)* (m:RtoX_b) and its inverse (m:XtoR_b) are used in the calculation of the *Decaying Component* (m:idc) of the short-circuit current. The *Equivalent Frequency fc/f (dc)* (m:fctof_b) shows the equivalent frequency ratio used for the decaying component calculation. For more information please refer to the *PowerFactory* User Manual or the IEC60909 standard.

Note: Both R/X ratios are typically calculated with an equivalent frequency and thus may differ from the R/X ratio obtained from the equivalent impedance.

The grid time constant, $\mathbf{T_N}$, is calculated as follows:

$$T_N = \frac{X}{R} \cdot \frac{1000}{2 \cdot \pi \cdot f_N} \tag{58}$$

where:

T_N is grid time constant in ms (*m:Tn*)

$\frac{X}{R}$ is peak X/R ratio (*m:XtoR*)

f_N is the nominal frequency of the network in Hz

3.1.1.3 Voltages

Table 3.3: Variables for voltages at terminals

Name	Unit	Description
ur	p.u.	Voltage, Real Part
ui	p.u.	Voltage, Imaginary Part
u	p.u.	Voltage, Magnitude
upc	p.u.	Voltage, Magnitude
u1	p.u.	Voltage, Magnitude
u1pc	%	Voltage, Magnitude
u1r	p.u.	Voltage, Magnitude
u1i	p.u.	Voltage, Magnitude
U	kV	Line-Ground Voltage, Magnitude
Ul	kV	Line-Line Voltage, Magnitude
phiu	deg	Voltage, Angle
du	%	Voltage Deviation
usr	p.u.	Transient Voltage, Real Part
usi	p.u.	Transient Voltage, Imaginary Part
us	p.u.	Transient Voltage, Magnitude
uspc	%	Transient Voltage, Magnitude
Us	kV	Transient Line-Ground Voltage, Magnitude
Uls	kV	Transient Line-Line Voltage, Magnitude
phius	p.u.	Transient Voltage, Angle
dus	%	Transient Voltage Deviation
urpref	p.u.	Pre-fault Voltage, Real Part
uipref	p.u.	Pre-fault Voltage, Imaginary Part
upref	p.u.	Pre-fault Voltage, Magnitude
uprefpc	%	Pre-fault Voltage, Magnitude
Upref	kV	Line-Ground Pre-fault Voltage, Magnitude
Ulpref	kV	Line-Line Pre-fault Voltage, Magnitude
phiupref	deg	Pre-fault Voltage, Angle

The calculated subtransient short-circuit voltage, $\underline{\mathbf{u}}''$, is available as:

$$\underline{u}'' = u_r'' + j \cdot u_i'' = u'' \cdot e^{j \cdot \varphi_{u''}} \tag{59}$$

where:

u_r'' is the real part of the subtransient short-circuit voltage in p.u. (*m:ur*)

u_i'' is the imaginary part of the subtransient short-circuit voltage in p.u. (*m:ui*)

u'' is the magnitude of the subtransient short-circuit voltage in p.u. (*m:u*)

$\varphi_{u''}$ is the phase angle of the subtransient short-circuit voltage in deg (*m:phiu*)

The positive sequence variables *m:u1*, *m:u1pc*, *m:u1r* and *m:u1i* are made available to to facilitate comparison between balanced and unbalanced calculations via *Comparison of Results*. They are calculated identically to their non positive sequence counterparts.

The angle between voltage and current, $\varphi_{\mathbf{u,i}}$, is calculated as follows:

$$\varphi_{u,i} = \varphi_{u''} - \varphi_k^{''} \tag{60}$$

where:

$\varphi_{u,i}$ is the angle between voltage and current in deg (*m:phiui*)

$\varphi_{u''}$ is the phase angle of the subtransient short-circuit voltage in deg (*m:phiu*)

$\varphi_k^{''}$ is the phase angle of the subtransient short-circuit current in deg (*m:phii*)

The calculated transient short-circuit voltage, $\underline{\mathbf{u}}^{'}$, is available as:

$$\underline{u}^{'} = u_r^{'} + j \cdot u_i^{'} = u^{'} \cdot e^{j \cdot \varphi_{u'}} \tag{61}$$

where:

$u_r^{'}$ is the real part of the transient short-circuit voltage in p.u. (*m:usr*)

$u_i^{'}$ is the imaginary part of the transient short-circuit voltage in p.u. (*m:usi*)

$u^{'}$ is the magnitude of the transient short-circuit voltage in p.u. (*m:us*)

$\varphi_{u'}$ is the phase angle of the short-circuit voltage in deg (*m:phius*)

The pre-fault voltage, $\underline{\mathbf{u}}_{\mathbf{p}}$, used for the calculation is available as:

$$\underline{u}_p = u_{p,r} + j \cdot u_{p,i} = u_p \cdot e^{j \cdot \varphi_{u_p}} \tag{62}$$

where:

$u_{p,r}$ is the real part of the pre-fault short-circuit voltage in p.u. (*m:urpref*)

$u_{p,i}$ is the imaginary part of the pre-fault short-circuit voltage in p.u. (*m:uipref*)

u_p is the magnitude of the pre-fault short-circuit voltage in p.u. (*m:upref*)

φ_{u_p} is the phase angle of the short-circuit voltage in deg (*m:phiupref*)

Note: The pre-fault voltage is obtained from the pre-fault load flow calculation or the voltage factor configured in the short-circuit command and does not include the additional voltage from converter driven plants.

All voltage variables are also available as denormalised quantities, $\mathbf{U}$ and $\mathbf{U_l}$, which are calculated as follows:

$$U = u \cdot \frac{U_N}{\sqrt{3}} \tag{63}$$

$$U_l = u \cdot U_N \tag{64}$$

where:

U is the magnitude of the line-ground voltage in kV (*m:U*, *m:Us*, *m:Upref*)

U_l is the magnitude of the line-line voltage in kV (*m:Ul*, *m:Uls*, *m:Ulpref*)

u is the magnitude of the voltage in p.u. (*m:u*, *m:us*, *m:upref*)

U_N is the nominal voltage of the terminal in kV (*e:uknom*)

Similarly, the convenience variables *m:upc*, *m:uspc* and *m:uprefpc* express the magnitude of the corresponding voltage in percent. The variables *m:du* and *m:dus* show the deviation of the corresponding short-circuit voltage from the nominal voltage in p.u..

3.1.1.4 Loadings

Table 3.4: Variables for loadings at terminals

Name	Unit	Description
Ithload	%	Loading, Thermal equivalent Current
Ipload	%	Loading, Peak Short-Circuit Current
Ithrtk	kA	Rated Short-Time Current (Tk)

Loading variables require rated values. These can be entered in the *Busbar Type* (TypBar) assigned to the terminal. If no type is assigned or the the corresponding rated value is zero, the loadings are assumed to be zero.

The thermal loading *m:Ithload* is calculated as follows:

$$I_{th,r}(T_k) = I_{th,r}^2 \cdot \frac{T_r}{T_k} \tag{65}$$

$$loading = \frac{I_{th}}{I_{th,r}(T_k)} \tag{66}$$

where:

$I_{th,r}$ is the rated short-time thermal current in kA (*t:Ithlim*)

T_r is the rated short-time in s (*t:Tkr*)

T_k is the *Fault Clearing Time (Ith)* (*Tk*) from the short-circuit command in s

$I_{th,r}(T_k)$ is the short-time thermal current rated to the fault clearing time in kA (*m:Ithrk*)

I_{th} is the calculated thermal equivalent short-circuit current in kA (*m:Ith*)

The peak short-circuit loading *m:lpload* is calculated as follows:

$$loading = \frac{i_p}{I_{p,r}} \tag{67}$$

where:

$I_{p,r}$ is the rated peak short-circuit current in kA (*t:lplim*)

i_p is the calculated peak short-circuit current in kA (*m:ip*)

3.1.2 Result Variables for Elements

Variables described in this section are available “per Connection”, unless stated otherwise. The corresponding designation has been omitted for readability.

3.1.2.1 Short-Circuit Current and Derived Values

Table 3.5: Variables for short-circuit current and derived values at elements

Name	Unit	Description
Ikss	kA	Initial Short-Circuit Current
ikss	p.u.	Initial Short-Circuit Current, Magnitude
phii	deg	Phase Current, Angle
Skss	MVA	Initial Short-Circuit Power
Iks	kA	Transient Short-Circuit Current
iks	p.u.	Transient Short-Circuit Current, Magnitude
phiiks	deg	Transient Short-Circuit Current, Angle
ip	kA	Peak Short-Circuit Current
Ib	kA	Short-Circuit Breaking Current
Sb	MVA	Short-Circuit Breaking Power
Ith	kA	Thermal equivalent Short-Circuit Current
I	kA	Short-Circuit Current
maxIkss	kA	Max. Initial Short-Circuit Current
maxIks	kA	Max. Transient Current
maxip	kA	Max. Peak Short-Circuit Current
maxIb	kA	Max. Breaking Current
maxIth	kA	Max. Thermal equivalent Short-Circuit Current
i1	p.u.	Positive-Sequence Current, Magnitude
I1	kA	Positive-Sequence Current, Magnitude
phii1	deg	Positive-Sequence Current, Angle
phiui	deg	Angle between Voltage and Current
phiu1i1	deg	Angle between Voltage and Current in Pos. Sequence System
cosphiui		cosphi

The calculated initial short-circuit contribution, $\underline{\mathbf{I}}''_{\mathbf{k,j}}$, is available as:

$$\underline{I}''_{k,j} = I''_{k,j} \cdot e^{j \cdot \varphi''_{k,j}} = i''_{k,j} \cdot I_r \cdot e^{j \cdot \varphi''_{k,j}} \tag{68}$$

where:

$I''_{k,j}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss* or *m:I*)

$\varphi''_{k,j}$ is the phase angle of the initial short-circuit contribution in deg (*m:phii*)

$i''_{k,j}$ is the magnitude of the initial short-circuit contribution in p.u. (*m:ikss*)

I_r is the rated current of the element in kA

The initial short-circuit power contribution, S''_k, is a fictitious value to facilitate comparison of short-circuit strengths across different voltage levels and is calculated as follows:

$$S''_{k,j} = \sqrt{3} \cdot I''_{k,j} \cdot U_N \tag{69}$$

where:

$S''_{k,j}$ is the initial short-circuit power contribution in MVA (*m:Skss*)

$I''_{k,j}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss* or *m:I*)

U_N is the nominal voltage in kV (*e:uknom* of the connected terminal)

The calculated transient short-circuit contribution, $\underline{\mathbf{I}}'_{\mathbf{k}}$, is available as:

$$\underline{I}'_{k,j} = I'_{k,j} \cdot e^{j \cdot \varphi'_{k,j}} = i'_{k,j} \cdot I_r \cdot e^{j \cdot \varphi'_{k,j}} \tag{70}$$

where:

$I'_{k,j}$ is the magnitude of the transient short-circuit contribution in kA (*m:Iks*)

$\varphi'_{k,j}$ is the phase angle of the transient short-circuit contribution in deg (*m:phiiks*)

$i'_{k,j}$ is the magnitude of the transient short-circuit contribution in p.u. (*m:iks*)

I_r is the rated current of the element in kA

The peak short-circuit current contribution, $\mathbf{i}_{\mathbf{p,j}}$, is calculated as follows:

$$i_{p,j} = \sqrt{2} \cdot \kappa \cdot I''_{j,rot} + \sqrt{2} \cdot I''_{j,cdp} \tag{71}$$

where:

$i_{p,j}$ is the peak short-circuit current contribution in kA (*m:ip*)

κ is a factor depending on the R/X ratio (*m:RtoX* at the short-circuit location)

$I''_{j,rot}$ is the rotating part of the initial short-circuit contribution in kA

$I''_{j,cdp}$ is the full size converter part of the initial short-circuit contribution in kA

The short-circuit breaking current contribution, $\mathbf{I_{b,j}}$, is calculated as follows:

$$I_{b,j} = \begin{cases} I_{k,j}^{'} + (I_{j,rot,break}^{''} - I_{j,rot}^{'}) \cdot e^{-25 \cdot T_b} & T_b < 120 \text{ ms} \\ I_{k,j}^{'} & T_b \geq 120 \text{ ms} \end{cases} \quad (72)$$

where:

$I_{b,j}$ is the short-circuit breaking current contribution in kA (*m:Ib*)

$I_{j,rot,break}^{''}$ is the rotating part of the initial short-circuit contribution excluding DFIGs in kA

$I_{j,rot}^{'}$ is the rotating part of the transient short-circuit contribution in kA

$I_{k,j}^{'}$ is the magnitude of the transient short-circuit contribution in kA (*m:Iks*)

T_b is the breaking time in s (*c:Tb* of the short-circuit location)

Similar to the initial short-circuit power, the short-circuit breaking power contribution, $\mathbf{S_{b,j}}$, is a fictitious value to facilitate comparison of short-circuit breaking requirements across different voltage levels and is calculated as follows:

$$S_{b,j} = \sqrt{3} \cdot I_{b,j} \cdot U_N \quad (73)$$

where:

$S_{b,j}$ is the short-circuit breaking power contribution in MVA (*m:Sb*)

$I_{b,j}$ is the short-circuit breaking current contribution in kA (*m:Ib*)

U_N is the nominal voltage in kV (*e:uknom* of the connected terminal)

The thermal equivalent short-circuit current contribution, $\mathbf{I_{th,j}}$ is a fictitious current expressing the thermal strain on equipment and is calculated as follows:

$$I_{th,j} = I_{k,j}^{''} \cdot \sqrt{m+n} \quad (74)$$

where:

$I_{th,j}$ is the thermal equivalent short-circuit current contribution in kA (*m:Ith*)

m is a factor expressing the heat effect of the DC component with

$$m = f(\kappa, T_k)$$

n is a factor expressing the heat effect of the AC component with

$$n = f\left(\frac{I_{k,j}^{''}}{I_{k,j}^{'}}, T_k\right)$$

$I_{k,j}^{''}$ is the magnitude of the initial short-circuit contribution in kA (*m:Ikss*)

$I_{k,j}^{'}$ is the magnitude of the transient short-circuit contribution in kA (*m:Iks*)

κ is a factor depending on the R/X ratio at the short-circuit location (*n:RtoX*)

T_k is the *Fault Clearing Time (Ith)* (*Tk*) from the short-circuit command in s

For a detailed description of the factors, $\mathbf{m}$ and $\mathbf{n}$, please refer to the IEC60909 [2, 3]. Due to the absence of a steady-state current, the *complete* method uses the transient short-circuit current (*m:Iks*) for calculating the factor n.

The calculation of the maximum short-circuit currents depends on the *Calculate max. Branch Currents = Busbar Currents* (iopt_brc) setting in the short-circuit command. If the setting is *enabled*, the values are taken directly from the short-circuit location. If the setting is *disabled*, the values for $I''_{k,max}$ and for $I'_{k,max}$ are calculated as follows :

$$I_{max} = |\max(\underline{I}_{bus} - \underline{I}_{branch}; \underline{I}_{branch})| \tag{75}$$

where:

$\underline{I}_{bus}$ is the complex short-circuit at the terminal in kA

$\underline{I}_{branch}$ is the complex short-circuit contribution of the branch in kA

The resulting values for $I''_{k,max}$, $I'_{k,max}$ and their corresponding partial rotating and full size converter parts are then used with the corresponding equations to calculate the derived values $i_{p,max}$, $I_{b,max}$ and $I_{th,max}$. Where required, the R/X ratio from the short-circuit location is used.

The angle between voltage and current, $\varphi_{\mathbf{ui}}$, is calculated as follows:

$$\varphi_{ui} = \varphi_u - \varphi_i \tag{76}$$

where:

φ_{ui} is the angle between voltage and current in deg (*m:phiui*)

φ_u is the subtransient voltage angle in deg (*m:phiu* of the connected terminal)

φ_i is the subtransient current angle in deg (*m:phii*)

The positive sequence variables *m:i1*, *m:I1*, *m:phii1*, *m:phiu1i1* are made available to to facilitate comparison between balanced and unbalanced calculations via *Comparison of Results*. They are calculated identically to their non positive sequence counterparts. The variable *m:cosphiui* is a convenience variable, showing the power factor for the calculated angle between voltage and current.

3.1.2.2 Impedances

Table 3.6: Variables for impedances at elements

Name	Unit	Description
R	Ω	Fault Impedance, Real Part
X	Ω	Fault Impedance, Imaginary Part
Z	Ω	Fault Impedance, Magnitude
phiz	deg	Fault Impedance, Angle
rSbase	p.u.	Fault Impedance, Real Part
xSbase	p.u.	Fault Impedance, Imaginary Part
zSbase	p.u.	Fault Impedance, Magnitude
XtoR		X/R ratio
RtoX		R/X ratio
Zs	Ω	Source Impedance, Magnitude
phizs	deg	Source Impedance, Angle
Rs	Ω	Source Impedance, Real Part
Xs	Ω	Source Impedance, Imaginary Part

The fault impedance, $\underline{\mathbf{Z}}$, is calculated as follows:

$$\underline{Z} = R + j \cdot X = Z \cdot e^{j \cdot \varphi_Z} = \frac{\underline{U}''}{\underline{I}_k''} \tag{77}$$

where:

R is the real part of the fault impedance in Ω (*m:R*)

X is the imaginary part of the fault impedance in Ω (*m:X*)

Z is the magnitude of the fault impedance in Ω (*m:Z*)

φ_Z is the angle of the fault impedance in deg (*m:phiz*)

$\underline{U}''$ is the complex subtransient voltage of the connected terminal in kV

$\underline{I}_k''$ is the complex subtransient current in kA

The convenience variables *m:XtoR* and *m:RtoX* show the X/R ratio and R/X ratio of the branch impedance.

Note: The fault impedance is back-calculated from calculation results. It therefore includes additional influences, such as pre-fault loading, and will not necessarily match input values.

The fault impedance is also available in per unit values (*m:rSbase*, *m:xSbase* and *m:zBase*). The corresponding base impedance, $\mathbf{Z_B}$, is calculated as follows:

$$Z_B = \frac{U_N^2}{S_{base}} \tag{78}$$

where:

S_{base} is the *Base Apparent Power* (Sbase) from the *Project Settings* in MVA

U_N is the nominal voltage in kV (*e:uknom* of the connected terminal)

The source impedance, $\underline{\mathbf{Z}}_\mathbf{s}$, is calculated as follows:

$$\underline{Z}_s = R_s + j \cdot X_s = Z_s \cdot e^{j \cdot \varphi_{Z_s}} = \frac{\underline{U}_p - \underline{U}''}{\underline{I}_k'' - \underline{I}_p} \tag{79}$$

where:

R_s is the real part of the source impedance in Ω (*m:Rs*)

X_s is the imaginary part of the source impedance in Ω (*m:Xs*)

Z_s is the magnitude of the source impedance in Ω (*m:Zs*)

φ_{Z_s} is the angle of the source impedance in deg (*m:phizs*)

$\underline{U}''$ is the complex subtransient voltage of the connected terminal in kV

$\underline{I}_k''$ is the complex subtransient current in kA

$\underline{U}_p$ is the complex pre-fault voltage of the connected terminal in kV

$\underline{I}_p$ is the complex pre-fault current in kA

3.1.2.3 Powers

Table 3.7: Variables for powers at elements

Name	Unit	Description
P	MW	Active Power
Q	Mvar	Reactive Power
i1P	p.u.	Positive-Sequence Active Current
i1Q	p.u.	Positive-Sequence Reactive Current
I1P	kA	Positive-Sequence Active Current
I1Q	kA	Positive-Sequence Reactive Current

The power flows, $\mathbf{P}$ and $\mathbf{Q}$, are calculated as follows:

$$P + j \cdot Q = \sqrt{3} \cdot \underline{U}^{''} \cdot \left(\underline{I}_k^{''}\right)^* \tag{80}$$

where:

P is active power in MW (*m:P*)

Q is reactive power in Mvar (*m:Q*)

$\underline{U}^{''}$ is the complex subtransient voltage of the connected terminal in kV

$\left(\underline{I}_k^{''}\right)^*$ is the complex conjugate of the subtransient current in kA

Note: The initial short-circuit power $S_k^{''}$ is a fictitious value and does not express the magnitude of the apparent power corresponding to P and Q.

The positive sequence active and reactive currents are calculated as follows:

$$\varphi = \angle\left(P + j \cdot Q\right)$$

$$I_{1,P} = i_{1,p} \cdot I_r \cdot \cos\varphi \tag{81}$$

$$I_{1,Q} = i_{1,q} \cdot I_r \cdot \sin\varphi \tag{82}$$

where:

φ is the angle of the apparent power

$I_{1,P}$ is the positive sequence active current in kA (*m:I1P*)

$I_{1,Q}$ is the positive sequence reactive current in kA (*m:I1Q*)

$i_{1,P}$ is the positive sequence active current in p.u. (*m:i1P*)

$i_{1,Q}$ is the positive sequence reactive current in p.u. (*m:i1Q*)

I_r is the rated current of the element in kA

Note: Close to the fault location the apparent power will be almost zero. The angle, and by extension the positive sequence active and reactive currents, will therefore be undefined.

3.1.2.4 Beaker Loadings

Table 3.8: Variables for breaker loadings at elements

Name	Unit	Description
BrkIpload	%	Breaker Loading (peak)
BrkIbload	%	Breaker Loading (breaking)
BrkmaxIb	kA	Max. Breaking Current (breaker time)

Loading variables require rated values. These can be entered in the *Switch Type* (TypSwitch) assigned to the breaker associated with the connection side. If no type is assigned or the the corresponding rated value is zero, the loadings are assumed to be zero.

The peak short-circuit loading *m:BrkIpload* is calculated as follows:

$$loading = \frac{i_{p,max}}{I_{p,r}} \tag{83}$$

where:

$I_{p,r}$ is the rated peak short-circuit current in kA (*t:Ipnom* of the breaker)

$i_{p,max}$ is the calculated maximum peak short-circuit current in kA (*m:maxip*)

The break short-circuit loading *m:BrkIbload* is calculated as follows:

$$loading = \frac{I_{b,max}}{I_{b,r}} \tag{84}$$

where:

$I_{b,r}$ is the rated breaking current in kA (*t:Ibnom* of the breaker)

$I_{b,max}$ is the calculated maximum breaking current in kA (*m:BrkmaxIb*)

The *Max. Breaking Current (breaker time)* (m:BrkmaxIb) is identical to the *Max. Breaking Current* (m:maxIb).

3.1.2.5 Protection Related

Table 3.9: Protection related variables at elements

Name	Unit	Description
Tfct	s	Fault Clearing Time

The fault clearing time, $\mathbf{T_{fct}}$ is calculated as follows:

$$T_{fct} = T_{trip} + T_{open} \tag{85}$$

where:

T_{fct} is the fault clearing time in s (*m:Tfct*)

T_{trip} is the calculated tripping time in s (*c:yout* of the relay (*ElmRelay*) or fuse (*RelFuse*))

T_{open} is the breaker opening time in s (*t:t_open* of the breaker)

If multiple protection operate the same breaker, the fastest tripping time is used. A value of 9999.999 s indicates that no trip occurred.

Note: The fault clearing time is only calculated if protection devices are considered in the short-circuit calculation.

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4 ANSI Short-Circuit

Short-circuit calculations according to ANSI consider three types of fault currents [4]:

- Momentary (or first cycle) currents: Relevant up to and including one cycle immediately following the occurrence of a fault;
- Interrupting currents: Applicable to MV and HV circuit breaker parting times. These pertain to the time period ranging from 1.5 to 4 cycles. AC decrement must be considered when calculating interrupting currents.
- 30-cycle (or steady-state) short-circuit currents: Relevant for the time period beyond the opening time of MV circuit breakers and falling within 30 cycles and beyond from the onset of the fault.

For a detailed explanation and derivation of the various quantities listed throughout this section, the reader is referred to [4].

It should be noted that for all fault currents calculated, the interrupting pre-fault voltage, U_{pref} is based on the rated busbar voltage, U_{rated}.
Note: If option *Consider Transformer Taps* is selected, the pre-fault voltage is recalculated.

4.1 Balanced Representation

4.1.1 Result Variables for Terminals

Table 4.1: Variables for momentary short-circuit current and derived values at terminals

Name	Unit	Description
$Isym_m$	kA	Sym. Momentary Current
$phii_m$	deg	Sym. Momentary Current, Angle
$Ssym_m$	MVA	Sym. Momentary Power
$I16asym_m$	kA	Asym. Momentary Current, Duty
$I27peak_m$	kA	Asym. Momentary Peak Current, Duty
$Iasym_m$	kA	Asym. Momentary Current, calculated
$Ipeak_m$	kA	Asym. Momentary Peak Current, calculated
$XtoR_m$		Momentary X/R Ratio
$Mfact_m$		Momentary Mult. Factor
R_m	Ω	Momentary Short-Circuit Impedance, Real Part
X_m	Ω	Momentary Short-Circuit Impedance, Imaginary Part
Z_m	Ω	Momentary Short-Circuit Impedance, Magnitude
$phiz_m$	deg	Momentary Short-Circuit Impedance, Angle
$phiui_m$	deg	Momentary Angle between Voltage and Current
$upref_m$	p.u.	Momentary Pre-fault Voltage
$Upref_m$	kV	Momentary Pre-fault Voltage

The Asym. Momentary Current is calculated according to either the simplified method, which uses a constant value of 1.6 under the assumption that $\frac{X}{R} = 25$ [4]:

$$I_{16asym_m} = 1.6 \cdot I_{sym_m} \tag{86}$$

or based on the actual X/R ratio itself:

$$I_{asym_m} = I_{sym_m} \cdot \sqrt{1 + 2 \cdot e^{\frac{-2 \cdot \pi}{(X/R)}}} = I_{sym_m} \cdot \sqrt{M_{fact_m}} \tag{87}$$

Similarly, the Asym. Momentary Peak Current is calculated according to either the simplified method, which uses a constant value of 2.7 under the assumption that $\frac{X}{R} = 25$ [4]:

$$I_{27peak_m} = 2.7 \cdot I_{sym_m} \tag{88}$$

or based on the actual X/R ratio itself:

$$I_{peak_m} = I_{sym_m} \cdot \sqrt{2} \cdot (1 + e^{\frac{-2 \cdot \pi \cdot \tau}{(X/R)}}) \tag{89}$$

where τ is calculated according to [4] (Eqn. (9.5)).

The Momentary Mult. Factor is calculated as follows:

$$M_{fact_m} = 1 + 2 \cdot e^{\frac{-2 \cdot \pi}{X/R}} \tag{90}$$

The Momentary Short-Circuit Impedance Magnitude is calculated from the complex impedance:

$$Z_m = |Z \cdot e^{j \cdot \phi_z}| = |R_m + j \cdot X_m| \tag{91}$$

The Momentary Angle between Voltage and Current is:

$$phiui_m = \angle \underline{U} - \angle \underline{I_{sym_m}} \tag{92}$$

where U is the voltage in kV.

Table 4.2: Variables for interrupting short-circuit current and derived values at terminals

Name	Unit	Description
$Ilvcb$	kA	LV Power Fuse Breaker
$Ilvfuse$	kA	LV Fuse
$Ilvmccbg20$	kA	Molded Case Circuit Breaker $>$ 20 kA
$Ilvmccb1020$	kA	Molded Case Circuit Breaker 10-20 kA
$Ilvmccb010$	kA	Molded Case Circuit Breaker 0-10 kA
$Isym_i$	kA	Sym. Interrupting Current
$phii_i$	deg	Sym. Interrupting Current, Angle
$Ssym_i$	MVA	Sym. Interrupting Power
$XtoR_i$		Interrupting X/R Ratio
$NACD$		No AC Current Decay
$Iremote$	kA	Remote Current Contribution
$phiIrem$	deg	Remote Current Contribution, Angle
R_i	Ω	Interrupting Short-Circuit Impedance, Real Part
X_i	Ω	Interrupting Short-Circuit Impedance, Imaginary Part
Z_i	Ω	Interrupting Short-Circuit Impedance, Magnitude
$phiz_i$	deg	Interrupting Short-Circuit Impedance, Angle
$phiui_i$	deg	Interrupting Angle between Voltage and Current
$upref_i$	p.u.	Interrupting Pre-fault Voltage
$Upref_i$	kV	Interrupting Pre-fault Voltage

The LV Power Fuse Breaker is calculated by default as follows:

$$I_{lv} = I_{sym_i} \tag{93}$$

and if $X/R(momentary) > (X/R)'$ according to:

$$I_{lv} = I_{sym_i} \cdot \frac{(1. + e^{\frac{-\pi}{(X/R)}})}{(1. + e^{\frac{-\pi}{(X/R)'}})} \tag{94}$$

where $(X/R)'$ is defined as follows:

- For LV Power Fuse Breaker: $(X/R)' = tan(arccos(0.15))$
- For LV Fuse: $(X/R)' = tan(arccos(0.2))$

For Molded Case Circuit Breakers the default calculation is:

$$I_{lvmccb_x} = I_{sym_i} \tag{95}$$

and if $X/R(momentary) > (X/R)'$:

$$I_{lvmccb_x} = I_{sym_i} \cdot \frac{(1. + e^{\frac{-\pi}{(X/R)}})}{(1. + e^{\frac{-\pi}{(X/R)'}})} \tag{96}$$

where:

- I_{lvmccb_x} is the molded case circuit breaker current in kA, where x signifies the current range considered (i.e. *m:Ilvmccbg20*, *m:Ilvmccbg1020* or *m:Ilvmccbg010*) according to the definition of $(X/R)'$:
  - For $>$ 20 kA (*m:Ilvmccbg20*): $(X/R)' = tan(arccos(0.2))$
  - For 10-20 kA (*m:Ilvmccbg1020*): $(X/R)' = tan(arccos(0.3))$
  - For 0-10 kA (*m:Ilvmccbg010*): $(X/R)' = tan(arccos(0.5))$

The NACD is calculated as:

$$NACD = \frac{I_{remote}}{I_{fault}} \tag{97}$$

The Interrupting Short-Circuit Impedance Magnitude and Interrupting Angle between Voltage and Current are calculated in a similar manner to Eqn. (91) and Eqn. (92), respectively.

Table 4.3: Variables for interrupting short-circuit current and derived values at terminals

Name	Unit	Description
$Isym2c$	kA	2 cycle Sym. Interrupting Current
$Isym3c$	kA	3 cycle Sym. Interrupting Current
$Isym5c$	kA	5 cycle Sym. Interrupting Current
$Isym8c$	kA	8 cycle Sym. Interrupting Current
$Mfsym2c$		2 cycle Int. Mult. Factor
$Mfsym3c$		3 cycle Int. Mult. Factor
$Mfsym5c$		5 cycle Int. Mult. Factor
$Mfsym8c$		8 cycle Int. Mult. Factor
$Itot2c$	kA	2 cycle Total Interrupting Current
$Itot3c$	kA	3 cycle Total Interrupting Current
$Itot5c$	kA	5 cycle Total Interrupting Current
$Itot8c$	kA	8 cycle Total Interrupting Current
$Mftot2c$		2 cycle Total Mult. Factor
$Mftot3c$		3 cycle Total Mult. Factor
$Mftot5c$		5 cycle Total Mult. Factor
$Mftot8c$		8 cycle Total Mult. Factor

The 2-,3-,5-, and 8-Cycle Symmetrical Interrupting Currents are calculated as follows:

$$I_{sym,xc} = M_{fsym,xc} \cdot I_{sym_i} \tag{98}$$

where:

- $I_{sym,xc}$ is the 2-,3-,5-, or 8-cycle symmetrical interrupting current in kA , where x is the number of cycles (*m:Isym2c*, *m:Isym3c*, *m:Isym5c* or *m:Isym8c*).
- $M_{fsym,xc}$ is the 2-,3-,5-, or 8-cycle interrupting multiplication factor (*m:Mfsym2c*, *m:Mfsym3c*, *m:Mfsym5c* or *m:Mfsym8c*) as given in the multiplying factor curves in [4].

The 2-,3-,5-, and 8-Cycle Total Interrupting Currents (*m:Itot2c*, *m:Itot3c*, *m:Itot5c* and *m:Itot8c*) are calculated in a similar manner.

Table 4.4: Variables for 30-cycle short-circuit current and derived values at terminals

Name	Unit	Description
$Isym_30$	kA	30 cycle Sym. Current
$phii_30$	deg	30 cycle Sym. Current, Angle
$Ssym_30$	MVA	30 cycle Sym. Power
R_30	Ω	30 cycle Short-Circuit Impedance, Real Part
X_30	Ω	30 cycle Short-Circuit Impedance, Imaginary Part
Z_30	Ω	30 cycle Short-Circuit Impedance, Magnitude
$phiz_30$	deg	30 cycle Short-Circuit Impedance, Angle
$phiui_30$	deg	30 cycle Angle between Voltage and Current
$upref_30$	p.u.	30 cycle Pre-fault Voltage
$Upref_30$	kV	30 cycle Pre-fault Voltage

The 30 cycle current, impedance and voltage values are analogous to the values calculated in Table 4.1.

Table 4.5: Additional variables for terminals

Name	Unit	Description
I	kA	Short-Circuit Current
$phii$	deg	Short-Circuit Current, Angle
$phiui$	deg	Angle between Voltage and Current
S	MVA	Short-Circuit Power
ur	p.u.	Voltage, Real Part
ui	p.u.	Voltage, Imaginary Part
u	p.u.	Voltage, Magnitude
upc	%	Voltage, Magnitude
$u1$	p.u.	Voltage, Magnitude
$u1pc$	%	Voltage, Magnitude
$u1r$	p.u.	Positive-Sequence Voltage, Real-Part
$u1i$	p.u.	Positive-Sequence Voltage, Imaginary-Part
U	kV	Line-Ground Voltage, Magnitude
Ul	kV	Line-Line Voltage, Magnitude
$phiu$	deg	Voltage, Angle
du	%	Voltage Deviation
$Ipload$	%	Loading, Peak Short-Circuit Current

The Loading, Peak Short-Circuit Current is:

$$I_{p_{load}} = \frac{I_{p_m}}{I_{p_{lim}}} \cdot 100 \tag{99}$$

where I_{p_m} is the larger of I_{27peak_m} and I_{peak_m}.

4.1.2 Result variables for Elements

Select result variables for single- and multiple-port elements following a balanced ANSI Short-Circuit calculation are presented in the following subsections.

Table 4.6: Variables for branch elements

Name	Unit	Description
$BrkIpload$	%	Breaker Loading (peak)
$BrkIbload$	%	Breaker Loading (Interrupting)
$BrkmaxIb$	kA	Max. Breaking Current (breaker time)

If the detailed definition of switches is available via user-defined switch types (*TypSwitch*), then the Breaker Loading (peak) is calculated as follows:

$$BrkI_{p_{load}} = \frac{maxI_{asym_m} \cdot 100}{(\frac{BrkI_{close}}{BrkK_{factor} \cdot BrkK'_{factor}})} \tag{100}$$

where:

- $maxIasym_m$ is the larger of I_{16asym_m} and I_{asym_m}
- $BrkI_{close}$ is the maximum closing and latching capability
- $BrkK_{factor}$ is the user-defined voltage range factor in the switch type
- $BrkK'_{factor}$ is the adjusted voltage range factor, recalculated as $BrkK'_{factor} = \frac{BrkU_{max}}{u_{bus}}$ if $\frac{BrkU_{max}}{u_{bus}} < BrkK_{factor}$.

$$BrkI_{b_{load}} = \frac{(BrkmaxI_b \cdot 100)}{(BrkI_{ratshc} \cdot BrkK'_{factor})} \tag{101}$$

where $BrkK'_{factor}$ is the adjusted voltage range factor, recalculated as $BrkK'_{factor} = \frac{BrkU_{max}}{u_{bus}}$ if $\frac{BrkU_{max}}{u_{bus}} < BrkK_{factor}$.

$$BrkmaxI_{b,x} = \frac{maxI_{tot_{x,c}}}{maxI_{sym_{x,c}}} \tag{102}$$

where $maxI_{tot_{x,c}}$ and $maxI_{sym_{x,c}}$ are the maximum total and symmetrical interrupting currents and x represents the number of cycles (i.e. 2, 3, 5 or 8).

4.2 Unbalanced Representation

For most variables, please refer to the definitions provided in Section 4.1.1.

Note: The ANSI unbalanced short-circuit calculation only considers 3-phase terminals (with or without neutral) and therefore does not provide results for 1- and 2-phase terminals.

Table 4.7: Variables for terminals

Name	Unit	Description
$I_{A,B,C}$	kA	Short-Circuit Current
$phii_{A,B,C}$	deg	Short-Circuit Current, Angle
$phiui_{A,B,C}$	deg	Angle between Voltage and Current
$S_{A,B,C}$	MVA	Short-Circuit Power
$ur_{A,B,C}$	p.u.	Voltage, Real Part
$ui_{A,B,C}$	p.u.	Voltage, Imaginary Part
$u_{A,B,C}$	p.u.	Voltage, Magnitude
$upc_{A,B,C}$	%	Voltage, Magnitude
$u1_{A,B,C}$	p.u.	Voltage, Magnitude
$u1pc_{A,B,C}$	%	Voltage, Magnitude
$u1r_{A,B,C}$	p.u.	Positive-Sequence Voltage, Real-Part
$u1i_{A,B,C}$	p.u.	Positive-Sequence Voltage, Imaginary-Part
$U_{A,B,C}$	kV	Line-Ground Voltage, Magnitude
$Ul_{A,B,C}$	kV	Line-Line Voltage, Magnitude
$phiu_{A,B,C}$	deg	Voltage, Angle
$du_{A,B,C}$	%	Voltage Deviation
$Ipload_{A,B,C}$	%	Loading, Peak Short-Circuit Current

All phase variables are calculated in a similar manner to those in Table 4.5.

Table 4.8: Sequence variables for terminals

Name	Unit	Description
$I0_i$	kA	Interrupting Zero-Sequence Current, Magnitude
$I1_i$	kA	Interrupting Positive-Sequence Current, Magnitude
$I2_i$	kA	Interrupting Negative-Sequence Current, Magnitude

Table 4.9: Sequence variables for terminals

Name	Unit	Description
$Z0_m$	Ω	Momentary Zero-Sequence Impedance, Magnitude
$Z1_m$	Ω	Momentary Positive-Sequence Impedance, Magnitude
$Z2_m$	Ω	Momentary Negative-Sequence Impedance, Magnitude

5 References

[1] Engineering Recommendation G74 Issue 2 Procedure to meet the requirements on IEC 60909 for the calculation of short-circuit currents in three-phase AC power systems. Standard, Energy Networks Association, 2020.

[2] Short-circuit currents in three-phase a.c. systems. Standard, International Electrotechnical Commission (IEC), 2001.

[3] Short-circuit currents in three-phase a.c. systems. Standard, International Electrotechnical Commission (IEC), 2016.

[4] IEEE violet Book: 551 IEEE Recommended Practice for Calculating Short-Circuit Currents in Industrial and Commercial Power Systems. Standard, IEEE, 2006.